

BS119

2026 Fall Semester
Practice Exercises

Chapter 16: Acid-Base Equilibria

AI may assist you, but it can fail you as well!

□ Use these values where needed

Periodic Table of the Elements

Main Group Representative Elements												Main Group Representative Elements					
1A ^a												8A					
1												18					
1												2					
1.00794												4.002602					
2												10					
6.941												20.1797					
11												13					
22.989770												26.981538					
19												31					
39.0983												69.723					
37												49					
85.4678												114.818					
55												81					
132.90545												204.3833					
87												113					
[223.02]												[284]					
88												114					
[226.03]												[289.2]					
103												115					
[262.11]												[288]					
104												116					
[261.11]												[293]					
105												117					
[262.11]												[294]					
106												118					
[266.12]												[294]					
107																	
[264.12]																	
108																	
[269.13]																	
109																	
[268.14]																	
110																	
[281.15]																	
111																	
[272.15]																	
112																	
[285]																	
113																	
[284]																	
114																	
[289.2]																	
115																	
[288]																	
116																	
[293]																	
117																	
[294]																	
118																	
[294]																	

Lanthanide series	57 La 138.9055	58 Ce 140.116	59 Pr 140.90765	60 Nd 144.24	61 Pm [145]	62 Sm 150.36	63 Eu 151.964	64 Gd 157.25	65 Tb 158.92534	66 Dy 162.50	67 Ho 164.93032	68 Er 167.259	69 Tm 168.93421	70 Yb 173.04
Actinide series	89 Ac [227.03]	90 Th 232.0381	91 Pa 231.03588	92 U 238.02891	93 Np [237.05]	94 Pu [244.06]	95 Am [243.06]	96 Cm [247.07]	97 Bk [247.07]	98 Cf [251.08]	99 Es [252.08]	100 Fm [257.10]	101 Md [258.10]	102 No [259.10]

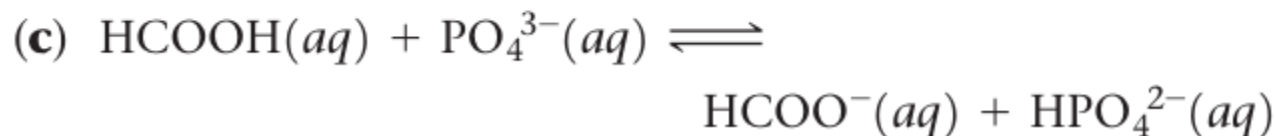
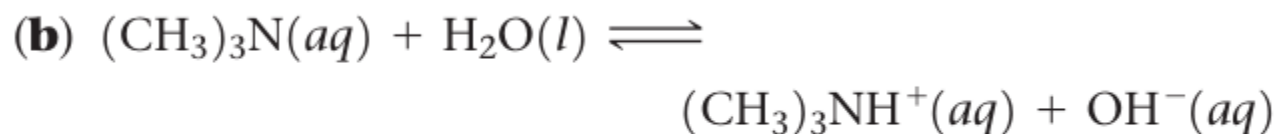
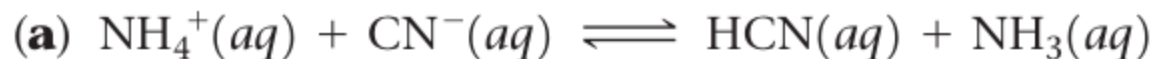
1. Define each of the following terms, phrases, laws, rules, etc. Give at least one example where needed.

- ✓ Acid-Base Equilibria
- ✓ Arrhenius acid
- ✓ Arrhenius base
- ✓ Brønsted-Lowry acid
- ✓ Brønsted-Lowry base
- ✓ Amphiprotic
- ✓ Conjugate acid-base pair
- ✓ Autoionization of Water
- ✓ Ion-product constant, K_w
- ✓ pH Scale
- ✓ Indicators
- ✓ Strong Acids
- ✓ Strong Bases
- ✓ Weak Acids
- ✓ K_a
- ✓ Weak Bases
- ✓ K_b
- ✓ Salts
- ✓ Hydrolysis
- ✓ Oxyacids
- ✓ Electronegativity
- ✓ Delocalized π bonding
- ✓ Lewis acid
- ✓ Lewis base

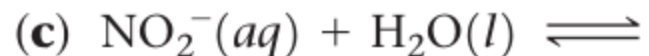
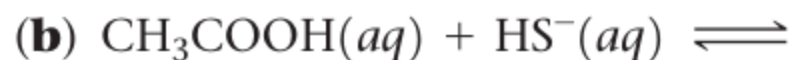
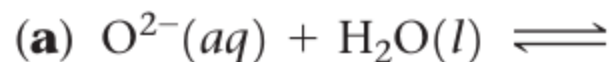
2. For each of these reactions, identify the acid and base among the reactants, and state if the acids and bases are Lewis, Arrhenius, and/or Brønsted–Lowry.



3. Identify the Brønsted–Lowry acid and the Brønsted–Lowry base on the left side of each of the following equations, and also identify the conjugate acid and conjugate base of each on the right side:



- 4.** Predict the products of the following acid–base reactions, and predict whether the equilibrium lies to the left or to the right of the reaction arrow.



- 5.** Calculate $[\text{H}^{+}]$ for each of the following solutions, and indicate whether the solution is acidic, basic, or neutral.

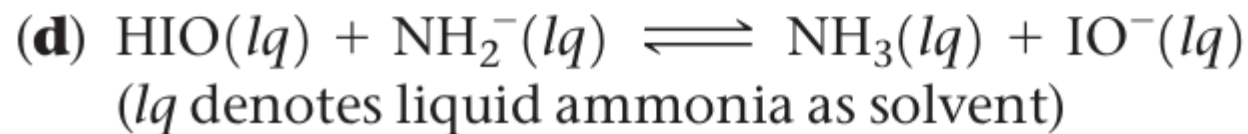
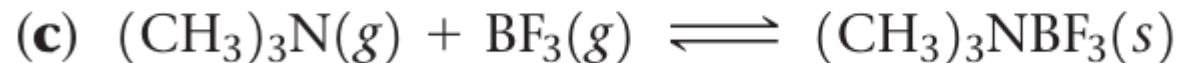
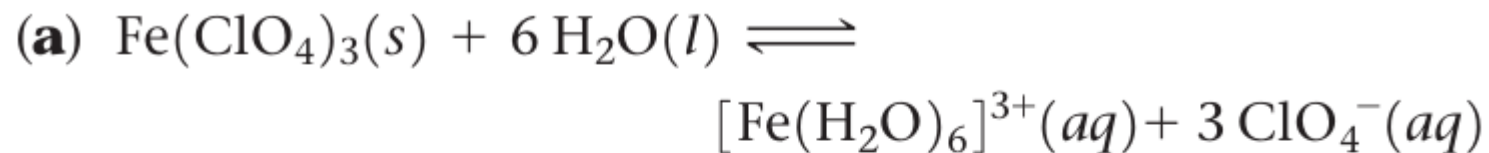
(a) $[\text{OH}^{-}] = 7.3 \times 10^{-10} \text{ M}$

(b) $[\text{OH}^{-}] = 0.015 \text{ M}$

(c) A solution in which $[\text{H}^{+}]$ is 10 times greater than $[\text{OH}^{-}]$.

- 6.** Write the chemical equation and the K_b expression for the reaction of each of the following bases with water:
- (a) Trimethylamine, $(\text{CH}_3)_3\text{N}$
 - (b) Sulfite, SO_3^{2-}
 - (c) Cyanide, CN^-
- 7.** Calculate the molar concentration of OH^- in a 0.050 M solution of ethylamine ($\text{C}_2\text{H}_5\text{NH}_2$; $K_b = 6.4 \times 10^{-4}$). Calculate the pH of this solution.
- 8.** A solution of potassium (KCH_3COO) has a pH of 8.90. What is the molarity of the solution?

- 9.** Predict whether aqueous solutions of the following compounds are acidic, basic, or neutral: **(a)** NH_4Br , **(b)** FeCl_3 , **(c)** Na_2CO_3 , **(d)** KClO_4 , **(e)** NaHC_2O_4 .
- 10.** Identify the Lewis acid and Lewis base among the reactants in each of the following reactions:



11. Indicate whether each of the following statements is correct or incorrect.

- (a) Every Brønsted–Lowry acid is also a Lewis acid.
- (b) Every Lewis acid is also a Brønsted–Lowry acid.
- (c) Conjugate acids of weak bases produce more acidic solutions than conjugate acids of strong bases.
- (d) K^+ ion is acidic in water because it causes hydrating water molecules to become more acidic.
- (e) The percent ionization of a weak acid in water increases as the concentration of acid decreases.

- 12.** A solution is made by adding 1.000 g $\text{Ca(OH)}_2(\text{s})$, 100.0 mL of 0.10 M H_2SO_4 , and enough water to make a final volume of 350.0 mL. Assuming that all of the solid dissolves, what is the pH of the final solution?
- 13.** Which, if any, of the following statements are true?
- (a) The stronger the base, the smaller the $\text{p}K_b$.
 - (b) The stronger the base, the larger the $\text{p}K_b$.
 - (c) The stronger the base, the smaller the K_b .
 - (d) The stronger the base, the larger the K_b .
 - (e) The stronger the base, the smaller the $\text{p}K_a$ of its conjugate acid.
 - (f) The stronger the base, the larger the $\text{p}K_a$ of its conjugate acid.

- 14.** The K_a of hypochlorous acid (HClO) is 3.0×10^{-8} at 25°C . What is the percent ionization of hypochlorous acid in a 0.015 M aqueous solution of HClO at 25°C ?
- 15.** Using the data in the table, which of the conjugate acids below is the weakest acid?

Base	K_b
NH_3	1.8×10^{-5}
$\text{C}_5\text{H}_5\text{N}$	1.7×10^{-9}
H_2NOH	1.1×10^{-8}
$\text{NH}_2\text{CH}_2\text{CH}_3$	4.2×10^{-4}

- A) NH_4^+
B) $\text{C}_5\text{H}_5\text{NH}^+$
C) H_3NOH^+
D) NH_3CH_3^+
E) NH_4^+ and NH_3CH_3^+

16. Using the data in the table, which of the conjugate acids below is the strongest acid?

Base	K_b
ClO^-	3.3×10^{-7}
CO_3^{2-}	1.8×10^{-4}
HS^-	1.8×10^{-7}
NH_2CH_3	4.4×10^{-4}

A) HClO

B) HCO_3^-

C) H_2S

D) NH_3CH_3^+

E) H_2S and HClO

17. The pH of a 0.25 M aqueous solution of hydrofluoric acid, HF, at 25.0 °C is 2.03. What is the value of K_a for HF?

- 18.** Among the given pairs of aqueous solutions, identify the weak acid that exhibits the lowest percentage ionization. Explain
- A. 0.01 M HCN ($K_a = 6.2 \times 10^{-10}$) **or** 0.01 M H_2CO_3 ($K_a = 4.5 \times 10^{-7}$)
- B. 0.01 M HNO_2 ($K_a = 4.0 \times 10^{-4}$) **or** 0.01 M HCHO_2 ($K_a = 1.8 \times 10^{-4}$)
- 19.** The acid-dissociation constants of phosphoric acid (H_3PO_4) are $K_{a1} = 7.5 \times 10^{-3}$, $K_{a2} = 6.2 \times 10^{-8}$ and $K_{a3} = 4.2 \times 10^{-13}$ at 25.0 °C. What is the pH of a 2.5 M aqueous solution of phosphoric acid?
- 20.** The pH of a 0.15 M aqueous solution of NaBrO (the sodium salt of HBrO) is 10.7. What is the K_a for HBrO?